Fixed-Index Wilson Majorant Weights and Longitudinal Edge Contributions

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Abstract

For the half-filled discrete prolate concentration matrix and its commuting finite Jacobi matrix, we prove an explicit cubic asymptotic for the positive Wilson-majorant expectation at every fixed extreme spectral index, with an order-four error bound. The proof combines the finite-Hermite corrected quasimode construction, Fourier concentration, and positivity of the majorant. Exact particle–hole and occupation–majorant relations, together with Slepian's fixed-index concentration asymptotic, determine the contribution of any fixed finite collection of extreme-mode pairs to the longitudinal reference-modular trace. No bound uniform in a growing spectral index is asserted.

1. Operators and conventions

Let $N = 2L$ be even. On $\mathbb{C}^N$, define the symmetric Jacobi matrix

$$(J_N)_{j,j+1} = \frac{(j+1)(N-j-1)}{N}, \qquad (J_N)_{jj} = 0, \quad 0 \le j < N-1.$$

The eigenvalues are simple. Denote its unit real eigenvectors in decreasing eigenvalue order by $v_{N,k}$, $0 \le k < N$. Let P_0 be the infinite-lattice Fourier projection onto $|p| \le \pi/2$, χ_N the projection onto the N-site interval, and $B_N = \chi_N P_0 \chi_N$. In Slepian's notation, $B_N = p(N, 1/4)$ and $J_N = \frac{2}{N}\sigma(N, 1/4)$. His §4.1, equations (129)–(131), pp. 1403–1404, identifies the descending Jacobi eigenvalue index with the descending concentration eigenvalue index. This ordering theorem is a separate classical input; it does not follow merely from commutation.

For $\mu > 0$, set $r_N = \mu/(N+\mu)$. In the interval-preserving scalarization, the full-line positive Wilson majorant has momentum multiplier

$$\widetilde{G}_r(p) = \frac{1}{2}\left(1 - \frac{|\cos p|}{\sqrt{\cos^2 p + r^2 \sin^2 p}}\right), \quad -\pi \le p \le \pi. \tag{1}$$

Write $G_N = \chi_N G_{r_N} \chi_N$ and

$$q_{N,k}(\mu) = \langle v_{N,k}, G_N v_{N,k} \rangle.$$

For every $r > 0$, $0 \le G_r \le I/2$. If $\hat{v}(p) = \sum_{j=0}^{N-1} v_j e^{ijp}$, our Fourier convention is

$$\langle v, G_N v \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} \widetilde{G}_{r_N}(p) |\hat{v}(p)|^2 \, dp. \tag{2}$$

2. Corrected Hermite approximation

Put $h = N^{-1/2}$, $y_j = h(j - (N-1)/2)$, and $H_N = NI/2 - J_N$. The limiting harmonic oscillator is

$$H = -\tfrac{1}{4}\partial_y^2 + 2y^2, \qquad H\psi_k = \varepsilon_k \psi_k, \quad \varepsilon_k = \frac{2k+1}{\sqrt{2}}.$$

For a fixed finite Hermite span, sampling $(\mathcal{S}_h u)_j = \sqrt{h}\, u(y_j)$ gives the lattice expansion

$$H_N \mathcal{S}_h u = \mathcal{S}_h(Hu + h^2 Pu + h^4 Qu) + O_u(h^6),$$

where

$$P = \partial_y(y^2\partial_y) + \tfrac{1}{2} - \tfrac{1}{48}\partial_y^4, \qquad Q = \tfrac{1}{12}\partial_y^2(y^2\partial_y^2) + \tfrac{1}{12}\partial_y^2 - \tfrac{1}{1440}\partial_y^6.$$

The finite Hermite corrections w_k, z_k, orthogonal to ψ_k, solve the successive off-diagonal perturbation equations

$$(H - \varepsilon_k)w_k = -(P - p_k)\psi_k,$$
$$(H - \varepsilon_k)z_k = -((P - p_k)w_k + (Q - \rho_k)\psi_k),$$

with $p_k = \langle \psi_k, P\psi_k\rangle$ and $\rho_k = \langle \psi_k, Q\psi_k\rangle + \langle \psi_k, Pw_k\rangle$. The solvability conditions are precisely the diagonal perturbative equations. After normalization, the sampled vector

$$u_{N,k} = \mathcal{S}_h(\psi_k + h^2 w_k + h^4 z_k)/\|\mathcal{S}_h(\psi_k + h^2 w_k + h^4 z_k)\|$$

obeys a spectral residual $O_k(h^6)$. The fixed-index spectral gap is bounded away from zero for sufficiently small h, by convergence to the simple oscillator eigenvalues. Therefore, with a suitable eigenvector sign,

$$\boxed{\|v_{N,k} - u_{N,k}\|_{\ell^2} = O_k(h^6).} \tag{3}$$

The construction and spectral isolation are established in *Fixed-Index Jacobi Spectral Expansion Through Inverse-Square Order*, §5; equation (3) is the consequence required here. It is not uniform when k grows with N.

3. Cubic expectation theorem

Theorem 1. For each fixed $k \geq 0$ and fixed $\mu > 0$,

$$\boxed{q_{N,k}(\mu) = \frac{\mu^2(2k+1)}{2\sqrt{2}\,N^3} + O_{\mu,k}(N^{-4}).} \tag{4}$$

Proof. Every $u_{N,k}$ is a normalized sample of a fixed finite Hermite combination, with coefficients expanding in even powers of h. Poisson summation and Schwartz decay imply Fourier concentration at $p = O(h)$, with superalgebraically small mass outside each fixed $(-\epsilon, \epsilon)$, $0 < \epsilon < \pi/2$. In particular,

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} p^2 |\hat{u}_{N,k}(p)|^2\, dp = h^2\|\psi_k'\|_2^2 + O_k(h^4),$$
$$\frac{1}{2\pi}\int_{-\pi}^{\pi} p^4 |\hat{u}_{N,k}(p)|^2\, dp = O_k(h^4). \tag{5}$$

On $|p| \leq \epsilon$, with $0 < r \leq 1$, the scalar multiplier has the uniform expansion

$$\widetilde{G}_r(p) = \frac{r^2 p^2}{4} + O_\epsilon(r^2 p^4 + r^4 p^4). \tag{6}$$

As $r_N = \mu h^2 + O_\mu(h^4)$, equations (2), (5), and (6) imply

$$\langle u_{N,k}, G_N u_{N,k}\rangle = \frac{\mu^2 h^6}{4}\|\psi_k'\|_2^2 + O_{\mu,k}(h^8). \tag{7}$$

The oscillator virial identity gives $\|\psi_k'\|_2^2 = \sqrt{2}(2k+1)$.

Positivity provides a sharper eigenvector-transfer estimate than the operator norm alone. From $\|G_N\| \leq 1/2$, equation (3), and (7),

$$\begin{aligned}|\langle v, G_N v\rangle - \langle u, G_N u\rangle| &\leq 2\|G_N^{1/2}u\|\,\|G_N^{1/2}(v-u)\| + \|G_N^{1/2}(v-u)\|^2 \\ &= O_{\mu,k}(h^9) + O_k(h^{12}).\end{aligned}$$

Together with (7) this gives an $O(h^8) = O(N^{-4})$ remainder and proves (4). □

The estimate depends on the third-order Hermite quasimode and remains a fixed-index theorem. A standard unweighted $\|v - u\|$ estimate alone would not provide the same conclusion without positivity.

4. Exact occupation bounds and edge-pair contribution

Let $S = G(I - 2P_0)$, $S_N = \chi_N S \chi_N$. For a unit interval-supported vector v, write $b = \langle v, P_0 v\rangle$, $q = \langle v, Gv\rangle$, $s = \langle v, Sv\rangle$. Since $[G, P_0] = 0$ and $0 \le 2G \le I$,

$$\boxed{0 \le q + s \le 1 - b, \qquad 0 \le q - s \le b.} \tag{8}$$

Indeed, $q + s = 2\langle (I - P_0)v, G(I - P_0)v\rangle$ and $q - s = 2\langle P_0 v, GP_0 v\rangle$.

Define $D = \operatorname{diag}((-1)^j)_{j=0}^{N-1}$. Directly from the half-filled kernel and the even-displacement majorant,

$$DJ_N D = -J_N, \quad DB_N D = I - B_N, \quad DG_N D = G_N, \quad DS_N D = -S_N. \tag{9}$$

If η is the odd reference scaling function, the self-adjoint discrepancy

$$E_N = \log(I - B_N) - \log B_N + N\eta(J_N/N)$$

also satisfies $DE_N D = -E_N$. Thus the bottom paired Jacobi vector has $b \mapsto 1 - b$, $q \mapsto q$, $s \mapsto -s$ and $e \mapsto -e$, with no ambiguity in the pair signs.

The classical Slepian fixed-index concentration asymptotic, with the discrete-prolate index correspondence, gives

$$1 - b_{N,k} = O_k(N^{k+1/2} e^{-2N \operatorname{arsinh} 1}) \quad \text{for fixed } k.$$

For each fixed k, the paired modular discrepancy has the finite limit

$$\lim_{N\to\infty} e_{N,k} = c_k = \tfrac{1}{2}\log\pi - \log(k!) - k\log 2 + (k + \tfrac{1}{2})[\log(2k+1) - 1]. \tag{10}$$

Theorem 2. For every fixed positive integer K, the top and bottom K pairs contribute to the original longitudinal trace by

$$\boxed{\sum_{k=0}^{K-1} (s_{N,k} e_{N,k} + s_{N,N-1-k} e_{N,N-1-k}) = -\frac{\mu^2}{\sqrt{2}\, N^3} \sum_{k=0}^{K-1} (2k+1) c_k + o_{\mu,K}(N^{-3}).} \tag{11}$$

Proof. From (8), $s_{N,k} = -q_{N,k} + O(1 - b_{N,k})$ at each fixed top index. The complementary occupations decay exponentially in N, so Theorem 1 and (10) imply $s_{N,k} e_{N,k} = -\mu^2 (2k + 1) c_k / (2\sqrt{2} N^3) + o(N^{-3})$. Equation (9) gives an identical contribution from the paired bottom index. Summing over the fixed set proves (11). □

5. Scope of the result

The fixed-index error constants may depend on k and μ, and no estimate is asserted for $K = K(N) \to \infty$. In particular, neither the fixed-index results nor the conformal boundary-energy bound establishes by itself the longitudinal estimate

$$\operatorname{Tr}(S_N E_N) = o_\mu(\log N).$$

The separate argument in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* establishes this longitudinal estimate with the stronger bound $O_\mu(1)$ and yields the leading entropy law. For the specified fixed-μ Wilson comparison, the finite part is Theorem 1 of *Finite Part of Wilson–Dirac Relative Entropy*. Corollary 4 of *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals* strengthens the longitudinal estimate to $O_\mu(N^{-1})$. Neither conclusion supplies uniform error bounds for the present fixed-index formulas with $K = K(N) \to \infty$. Numerical tests are supplementary evidence, not interval-certified enclosures or proof of moving-index uniformity.

References

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5. K. Justus, *Exact Spectral Majorants for a Wilson–Dirac Interval* (2026), for the interval-preserving scalarization and full-line inequalities.

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